



Phonetic trace effects in experimentally-induced vowel errors

Manasvi Chaturvedi & Jason Anthony Shaw

To cite this article: Manasvi Chaturvedi & Jason Anthony Shaw (2025) Phonetic trace effects in experimentally-induced vowel errors, Cognitive Neuropsychology, 42:3-4, 106-126, DOI: [10.1080/02643294.2025.2573172](https://doi.org/10.1080/02643294.2025.2573172)

To link to this article: <https://doi.org/10.1080/02643294.2025.2573172>



Published online: 27 Oct 2025.



Submit your article to this journal [↗](#)



Article views: 185



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)



Phonetic trace effects in experimentally-induced vowel errors

Manasvi Chaturvedi  and Jason Anthony Shaw

Department of Linguistics, Yale University, New Haven, CT, USA

ABSTRACT

Speech errors in typical and atypical populations provide an important empirical foundation for theories of speech production and their neural bases. Studies of consonant speech errors have provided important evidence for cascading activation, c.f. categorical selection, across linguistic representations. Specifically, consonant errors show a phonetic trace effect whereby the phonetic details of the errorful consonant retain an influence of the intended sound. In this paper, we extend an experimental paradigm for speech error elicitation to vowels. Results show that vowel errors are similar to consonants in that errors differ systematically from their canonical (non-error) counterparts on the dimensions of primary control in articulation, as indexed by formant measures. We also found that vowel errors were not consistently different from canonical productions on the dimension of duration. We discuss the implications of these results for models that implement cascading activation to relate speech output to higher levels of language production.

ARTICLE HISTORY

Received 6 January 2025
Revised 26 September 2025
Accepted 30 September 2025

KEYWORDS

Speech errors; vowels;
cascading activation;
phonological competition;
speech planning; speech
production

1. Introduction

Language production is assumed to progress from higher-level representations—e.g., conceptual, syntactic, phonological—to articulator movements and the resulting speech sounds. Language production models aim to characterize this process, addressing issues such as competition and selection at each level of representation and the types of interactions between and within levels (e.g., Dell, 1986; Fromkin, 1971) (see Levelt, 1999 for a review).

The study of speech errors provides an important empirical foundation for models of language production (Levelt, 1999). First, speech errors can reveal the organization of different levels of planning—for example, as Levelt (1999) (p. 225) points out, high-frequency “mixed” errors, where the error is both *semantically* and *phonologically* related to the intention have provided an important basis for model proposals that involve interactions between phonological and semantic planning levels that go both ways, i.e., are “bi-directional” (e.g., Dell, 1986). As another example, the phonological *well-formedness* of speech errors, observed in early transcription studies of speech errors, has been interpreted as indicating a deep connection between tacit grammatical knowledge and speech production processes (Fromkin, 1971). Speech errors themselves are not random deviations from intentions; errors are also patterned, and therefore provide a unique

channel of information about the speech production process—a long-standing observation in speech error research (Fromkin, 1971; Garrett, 1975; Meringer & Mayer, 1895). Mowrey and MacKay (1990) state the rationale as follows: “speech errors result from transient malfunctions of the system by which utterances are produced [...] such malfunctions are consistent with, and thus may be interpreted as revealing, aspects of the production system” (p. 1). For example, speech errors can provide evidence about which linguistic units interact with one another and over what temporal intervals. The modelling of errors proceeds by considering interactions between linguistic representations and proposing how the selection of such representations progresses over time. Thus, in this way, modelling errors amounts to modelling language production itself.

Early models of speech planning posited a “categorical selection” process, where speech sounds are selected in higher levels of planning before being articulated (Fromkin, 1971; Levelt, 1992). The prediction arising from such categorical selection-based models is that competitors in higher levels of planning have no effect on the phonetic details of speech—since the competitors are not selected, they cannot influence subsequent levels of speech. Accordingly, for the case of speech errors, an error based on categorical mis-selection should be equivalent in phonetic detail to its canonical (non-error) counterpart. For example, if a speaker

produces *bath* when they had intended to say *path*, the [b] in the errorful production would be equivalent to a [b] in a canonical production, i.e., when the [b] is actually intended. This prediction has been tested in laboratory experiments that induce speech errors using fast repetition of tongue twisters or word pairs made up of nonce or real words (e.g., Frisch & Wright, 2002; Goldrick & Blumstein, 2006; Goldstein et al., 2007; Mowrey & MacKay, 1990), the SLIP task (Pouplier, 2007), and also via the study of speech errors in natural corpora (Alderete et al., 2021). In both types of data (experimental and corpus), the prediction of the “categorical selection model” is not borne out. What we see instead are effects of higher level phonological competition on the phonetics. Phonetic measurements of errorful speech show that there remains a phonetic trace of the intended speech category, even in errors that are perceived as categorical (e.g., Goldrick & Blumstein, 2006). Following past research, we refer to this as a “trace effect.” The trace effect in speech errors has been taken as evidence that phonological competitors (for errors, these are the intended and errorful sounds) can be co-activated, and that co-activation “cascades” down to articulatory processes, i.e., there is “cascading activation” across levels of representation (Goldrick, 2006; Peterson & Savoy, 1998). In contrast to the “categorical selection” account, each level of planning does not necessarily select just one representation to pass onto the next level of processing (Goldrick & Blumstein, 2006).

While trace effects have been established for consonants, vowels have not been investigated in the same way. The evidence for phonetic traces of the intention in speech errors, which constitutes strong evidence against categorical selection, has come largely from consonants. This may be because vowel errors in general have been studied less (Shattuck-Hufnagel, 1986) (notable exceptions include Laver, 1980; Rusaw & Cole, 2011; Shattuck-Hufnagel, 1986), and are reported as less frequent, based on both lab and corpus studies (Dell et al., 2000; Rusaw & Cole, 2011; Shattuck-Hufnagel, 1986). In fact, Rusaw and Cole (2011) found that vowel errors generally occurred in conjunction with consonant errors, while consonant errors often occurred alone. Vowels and consonants also show other differences in processing—such as activated brain regions, differences in categorization and discrimination, and mutability (e.g., Boatman et al., 1997; Caramazza et al., 2000; Eimas, 1963; Fry et al., 1962; Stevens et al., 1969; Van Ooijen, 1996). However, a difference in reported error frequency between vowels and consonants does not tell us anything about whether there might be a trace effect in vowels. Given the processing differences just mentioned, we cannot assume that we will see a trace

effect in vowels of the kind that we see in consonants. Vowel-consonant differences and error findings in both thus point to two gaps in our understanding of speech planning: (1) Whether vowel errors will be as frequent as consonant errors in tasks similar to those used to elicit consonant errors, and (2) Whether vowel errors, if any, will show trace effects and what the nature of such trace effects will be. In this paper, we aim to address both of these gaps.

The behaviour of vowel errors is important also because it can adjudicate between current speech production models that implement “cascading activation.” Stern et al. (2022)’s dynamic neural field (DNF) model and Goldrick and Chu (2014)’s gradient symbolic computation (GSC) and gesture duration model both account for consonant trace effects. They share the premise that trace effects result from the co-activation of errorful and intended representations, but they derive the resulting *phonetic* effects in different ways. These models make distinct predictions for trace effects in vowels, an issue we return to in Section 1.2.1. As we will see, the former predicts trace effects in vowel formants of errorful vowels, while the latter predicts shorter duration for errorful vowels. The current study, which experimentally elicits vowel errors, will thus help differentiate between these models.

In the remainder of this section, we will discuss an important issue in the study of speech errors—how to detect competitor effects in the phonetic signal (Section 1.1)—and then move on to the aforementioned models (Section 1.2). We will then describe our experiment, which generalizes a paradigm used in past studies of consonant errors to elicit vowel errors. Following description of the methods (Section 2) and results (Section 3), we discuss implications for the speech production models, pointing out common neural mechanisms linking experimentally-induced speech errors to speech deficits in clinical populations (Section 4).

1.1. Detecting competitor effects in the phonetic signal

Trace effects are a kind of competitor effect in that they reveal cognitive interactions, i.e., competition and selection, between higher-level categories, here, sound categories, which could be abstract phonemes or articulatory gestures (Goldstein et al., 2007). Other types of competitor effects observed in phonetic measures include dissimilation effects such as hyperarticulation in the presence of phonological competitors (e.g., Baese-Berk & Goldrick, 2009; Stern & Shaw, 2023; Wedel et al., 2018) and in a cue-target task, when the cue and target are different phonological categories

(Tilsen, 2009). Such competitor effects in the acoustic signal may not always be large enough to be noticeable by the human ear (as noted by Pouplier, 2007 for their intrusion errors, discussed below). Detecting competitor effects is an important general consideration and one that has been addressed in a variety of ways across studies of speech errors.

Within studies of consonant errors, phonetic trace effects have been seen in both acoustic and articulatory measurements of speech. An example of an acoustic measure showing trace effects is voice onset time (VOT), the time between the release of the stop closure and the onset of voicing (Lisker & Abramson, 1964). This phonetic dimension is contrastive for English, specifying the difference between voiced (lower VOT) and voiceless (higher VOT) consonants (Lisker & Abramson, 1964), e.g., [k] vs. [g]. Articulatory trace effect evidence has come from measures such as tongue height (for different parts of the tongue, like tongue tip or tongue dorsum) (e.g., Goldstein et al., 2007; Pouplier, 2007). Below, we review in more detail key studies that have contributed to this literature and the methods used to detect trace effects in each.

In a task involving the production of tongue twisters of types ABBA and BAAB, where A and B were stop-initial items like *keff* or *geff*, Goldrick and Blumstein (2006) found that the VOT of errorful productions was different from their canonical counterparts. For example, if *geff* was incorrectly produced instead of *keff* in a tongue twister like *keff geff geff keff*, the VOT of the errorful [g] was higher than its canonical [g] counterpart—i.e., the VOT showed the influence of /k/, a voiceless stop, which in general have longer VOT than voiced stops (Chodroff & Wilson, 2017; Lisker & Abramson, 1964). Correspondingly, errorful voiceless stops (intended voiced, and produced voiceless) had shorter VOT than their canonical counterparts—which, again, shows the influence of the intention, the voiced /g/. In Goldrick and Blumstein (2006)'s study, the decision to classify each token as an error or non-error was done perceptually. Trace effects were evaluated by comparing the VOT of errors to non-errors in the same environment.

Extending this work, Alderete et al. (2021) carried out a comparison of speech errors in stops in natural speech and in lab studies similar to Goldrick & Blumstein's (specifically, Goldrick et al. 2011, 2016), finding that the trace effect was present in both kinds of data. Additionally, they used two methods of error classification—perception-based classification for the corpus tokens and a statistical method following Goldrick et al. (2011) (described below) for the experimental data. Goldrick et al. (2011) classified tokens as voiced or voiceless

using a mixture model trained on each participant's VOT distribution and the maximum likelihood decision threshold—i.e., the boundary above or below which a token was likely to be voiceless or voiced. If this classification matched the intention (what would be the “correct” phoneme), the token was labelled a non-error. If the intention and classification did not match, it was an error. Trace effects were then determined by comparing the VOT of errors and non-errors in closely matched prosodic contexts.

In addition to the acoustic dimension of VOT, error-intention interactions have also been found in articulatory measures (Goldstein et al., 2007; Pouplier, 2007). Pouplier (2007) and Goldstein et al. (2007) collected electromagnetic articulography (EMA) data on the tongue tip and tongue dorsum trajectories of stop consonants. They found errors of two types: intrusion errors, the presence of an intruding gesture that should not be present, and reduction errors, which correspond to errorful gestures that are reduced in magnitude as compared to their canonical counterparts. For example, for a [t] production, an intrusion would be associated with a tongue dorsum gesture, while a reduction would be associated with a tongue tip gesture of lower amplitude than canonical [t] (Pouplier, 2007, p. 321). Error classification was done differently in the two studies. Pouplier (2007) classified errors based on distributions of tongue dorsum and tongue tip maximum amplitude—specifically, “individual datapoints that rendered these otherwise distinct distributions overlapping were considered to be errors” (Pouplier, 2007, p. 322). Goldstein et al. (2007), on the other hand, considered all productions that lied outside of two standard deviations of the mean tongue dorsum/tongue tip height (based on the canonical utterances) to be instances of errors. Pouplier (2007) compared errorful and non-errorful productions matched for phonological context, while Goldstein et al. (2007) compared errorful productions coming from contexts with alternating consonants to control productions with no competitor influence.

The studies discussed above show, via various error classification methods, that competitor effects can be detected in lower-level phonetic information, such as VOT and tongue height. However, the detection of competitor effects does not require error classification. Indeed, McMillan and Corley (2010) comment that the notion of an “error,” and thus the classification method chosen, may be arbitrary: “assigning articulations to categories such as “error” [...] imposes an essentially arbitrary threshold below which the influence of competing phonemes is not considered” (p. 246). McMillan and Corley (2010) chose instead to detect competitor effects via a measure of articulatory variability. They

found more articulatory variability in the presence of overt phonological competitors (ABBA trials, where A and B represent different stop consonants) than a control condition lacking competitors (AAAA trials, where A represents the same stop consonant). This particular measure effectively detects competitor effects without error classification. However, in addition to the presence/absence of competitor effects, the direction of the effect can be revealing as to the mechanism behind it. For example, as noted before, the trace effects discussed above are assimilatory but competition effects can also result in dissimilation (Beach & Niziolek, 2024; Tilsen, 2009; Wedel et al., 2018). Discerning the direction of the effect—assimilatory vs. dissimilatory—may be important for pinpointing the causal

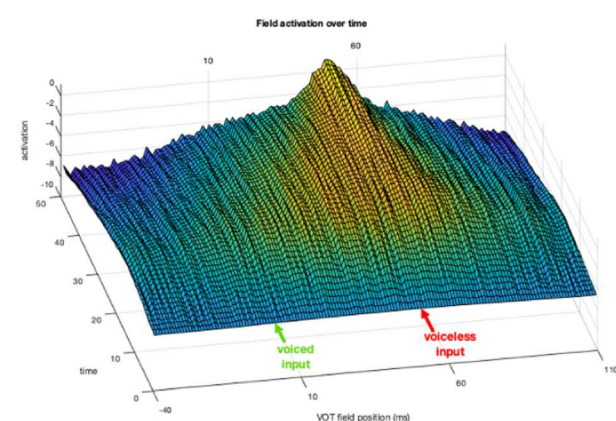
mechanism, an issue we return to below in our discussion of speech production models.

In order to detect assimilatory trace effects via comparison of errors and non-errors, we took a hybrid approach, comparing tokens in control vs. critical trials (Goldstein et al., 2007; McMillan & Corley, 2010) while adopting a statistical method for classification of errors. An important consideration for our study is vowel-to-vowel coarticulation, the phenomenon whereby the phonetic properties of a vowel are influenced by temporally adjacent vowels (Magen, 1997; Öhman, 1966). Vowel-to-vowel coarticulation is assimilatory, reflecting the temporal overlap of articulatory gestures, and has been argued to emerge in planning (Whalen, 1990). In order to discern trace effects from vowel-to-vowel coarticulation, our stimulus design mirrors McMillan and Corley (2010)'s AAAA/ABBA design, with initial vowels representing the “A” and “B” tokens instead of stops. This allows us to compare vowels with (ABBA) and without (AAAA) possible vowel-to-vowel coarticulation. Our classification method is a Support Vector Machine (SVM), trained on the control condition (AAAA) and tested on the alternating condition (ABBA).

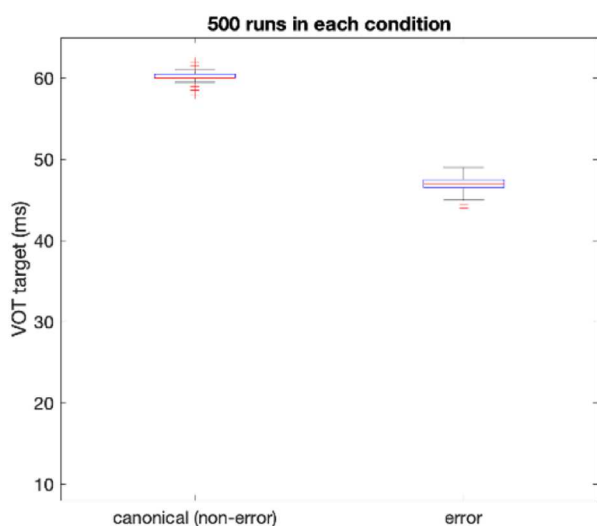
Before moving on to the methods of our experiment, we first describe in greater detail the existing models of trace effects, developed on the basis of consonant errors, and their predictions for vowels.

1.2. Models of trace effects

Stern et al. (2022) propose a model that captures VOT trace effects using the framework of dynamic field theory (DFT) (Schöner & Spencer, 2016). DFT models perception and action behaviour using dynamic neural fields (DNFs) and the evolution and decay of activation in these DNFs over time. DNFs consist of populations of neurons sensitive to a particular sensory or action dimension. For speech, this may be VOT, tongue position, or other acoustic or articulatory dimensions of speech. Inputs into neural fields, which take a Gaussian shape, represent percepts or action intentions. Inputs induce changes in the dynamic attractor of the fields. Activation in a DNF evolves over time towards stable attractor states. Cognitive decisions are represented by stable states in DNFs, i.e., stabilized activation peaks over particular field values. The DNF parameters can ensure that only one stabilized peak forms, even in the presence of multiple inputs to the field. Thus, the field can be described as “selecting” between inputs—a “selection dynamics.” However, since all inputs are represented in a continuous space, the option for categorical selection coexists with a compromise, whereby



(a)



(b)

Figure 1. Stern et al. (2022) results for voiceless errors showing trace for the voiced intention. (a) One simulation and (b) 500 simulations.

inputs may interact to cause an activation peak that is intermediate between input values. The interaction kernel plays an important role here. By specifying that neurons close to each other will excite each other and neurons far away from each other will inhibit each other, active neurons and neurons close to them grow further in activation while neurons more distal to an activation peak are inhibited, leading to stabilization of the activation peak.

The trace effect in Stern et al. (2022)'s model is captured via *overlapping* intention and competitor inputs into a VOT planning field. The presence of these two overlapping inputs leads to the stabilization of an activation peak at a field location that is between the two inputs, closer to the error, but pulled towards the intention (the latter is ensured by setting the error input to have a higher amplitude than the intention input). Below are two figures showing simulation results for voiceless errors (intended voiced). Figure 1(a) is the end result of one simulation run, i.e., the evolution of the field dynamics from rest to activation peak. We can see that with one input centered on a VOT of 60 ms (voiceless stop input, errorful with higher activation), and one centered on 10 ms (voiced stop input, intention with lower activation), we get stabilization around 45 ms—giving us an errorful voiceless stop with a trace of the voiced intention. Note that a key assumption here is that cascading activation from higher level representations leads to higher amplitude input at the voiceless (errorful) end of the DNF than the voiced end. Representing categories as continuously valued inputs into a DNF allows them to interact, giving rise to the trace effect, under certain conditions. Figure 1(b) shows the distribution of stabilized activation peaks over 500 simulation runs, comparing the location of the stabilized activation peak with two inputs (error and intention) to the location of the peak with only one input (the intention).

Goldrick and Chu (2014)'s model of trace effects makes use of gradient symbolic computation, or GSC (from Smolensky et al., 2014). GSC posits the idea of representations that are both symbolic and gradient—the former because representations are made up of symbols (for example, distinctive features) and the latter because symbols are represented as vectors of activation values. The activation values themselves are gradient and therefore can represent a blend of symbols. The ability to represent blends of symbols in a gradient activation vector makes the framework particularly well-suited for modelling the co-activation of competitors and targets, as observed in speech errors.

The process through which symbolic representations can come to be blended in gradient activation vectors is as follows. Activation vectors are driven over time

towards symbolic states, i.e., the vector activation pattern of a pure symbol, under the influence of two forces: optimization and quantization.

Optimization, implemented through stochastic gradient ascent, pulls the system towards the vector activation pattern of the optimal pure symbol. Quantization pulls the system towards (pure) symbolic states, without regard for optimality. Given enough time, optimization will dominate and activation patterns will settle on the optimal pure symbol (Smolensky et al., 2014). Under time pressure, quantization can drive the system toward a local, sub-optimal state or a blend of the local and optimal activation patterns, the process hypothesized to underlie speech errors and the trace effect (Goldrick & Chu, 2014; Smolensky et al., 2014). For example, the activation of the [∅ voiceless] feature will be lower (and the [voiced] feature will be higher) for an errorful *tore* production than a canonical *tore* production, meaning that the errorful *tore* would have a dominant influence of a voiceless stop and a trace of the voiced stop (Goldrick & Chu, 2014, p. 453).

How gradient activation representing multiple impure symbolic states will influence the phonetic signal is outside of the immediate scope of GSC. Goldrick and Chu (2014) propose that the gradient activations of features map to gesture durations, such that lower activations, i.e., vector activations that are further from a pure state, result in shorter gestures and higher activation results in longer gestures. They demonstrate that this assumption derives trace effects in available consonant data, including VOT and tongue height. VOT for errorful voiceless stops is shorter because the activation of the [∅ voiceless] feature is lower (and the activation of the [voiced] feature is higher), a consequence of blending with the voiced competitor. Tongue tip height is lower in errorful stops because, on their account, there is insufficient time to achieve the tongue tip target, i.e., target undershoot (Browman, 1990; Lindblom, 1963; Parrell & Narayanan, 2018). Thus, the trace effect in consonant errors is accounted for by assuming a mapping from activation to gesture duration, although it has also been noted that this assumption may not generalize to other cases of errors, including other types of consonant errors (Pouplier & Goldstein, 2014).

1.2.1. Predictions

The two models proposed to account for consonant trace effects make specific predictions for vowel errors.

Let us begin with the second model we discussed, the GSC and gesture duration model (Goldrick & Chu, 2014). The most obvious prediction of this model is that errorful vowels should be shorter than their non-errorful

counterparts. This follows from the prediction that errors, in general, have shorter gestures than non-errors, a prediction that can be easily tested. Additionally, shorter gesture durations may in turn lead to formant undershoot in vowels (Lindblom, 1963; Moon & Lindblom, 1994).

On the other hand, while Stern et al. (2022)'s model is based on a DNF where the metric dimension is VOT, we can generalize this model to vowels by considering the phonetic dimensions relevant to vowel production. Keeping in mind that DNFs should be based on dimensions that are under cognitive control in speech, we render predictions for vowels using the same mechanism as Stern et al. (2022) (overlapping inputs) for a DNF where the metric dimension is tongue position. Specifically, we define our DNF for vowels as tongue body position relative to the pharynx, or constriction location (CL). This dimension allows us to differentiate front vowels, which have a relatively large distance of the tongue body to the pharynx, from back vowels, which have a relatively short distance from tongue body to pharynx.

Importantly, there are fairly direct relations between the position of the tongue body and acoustic measures of vowel quality. The anterior position of the tongue body influences the resonance properties of the vocal tract, which can be detected in a spectral analysis of the acoustic signal. Specifically, for relatively low vowels, the cavity of the vocal tract in front of the tongue body and behind the tongue body each contribute a peak (high amplitude at a particular frequency) to the acoustic spectrum. These spectral peaks are typically referred to as formants—the lowest frequency formant is F1 and the next lowest is F2. For many vowels, F1 corresponds to the height of the tongue body while F2

corresponds to anteriority. We assume that the position of the tongue body is under control for vowel production, an articulatory dimension that correlates closely with vowel formants (Blackwood Ximenes et al., 2017).

Having established the metric dimension of the relevant DNF, we can now show how the mechanism deriving the trace effect can be generalized. Consider the vowel pair æ - ɑ. With two overlapping inputs into this planning field, we get a final peak location that is between them (Figure 2)—this intermediate constriction location will correspond to a change in the *formants* of the resulting vowel. If the constriction location is unlike the canonical version of the vowel, and pulled towards the competitor (the intention), we will see this pull in the formant measures as well. We should thus see a trace effect of the kind discussed for VOT: errorful vowel formant values that are pulled towards the formant values of the intention. For æ - ɑ, this would mean that errorful [ɑ]s, for example, will have a slightly higher F2 as compared to canonical [ɑ]s—i.e., they will be slightly æ-like. Note that this is a different prediction than Goldrick and Chu (2014)'s model, which could only predict lower F2, as a consequence of formant undershoot, driven by shorter vowel duration.

To summarize, the two models discussed above make specific predictions about vowel errors: one predicts that the duration of errorful vowels will be shorter (possibly with formant undershoot), and the other that formant measures will show a trace effect. Below we report the results of a tongue twister experiment—modelled on Goldrick and Blumstein (2006)'s and McMillan and Corley (2010)'s consonant error studies—that provides a test of these predictions.

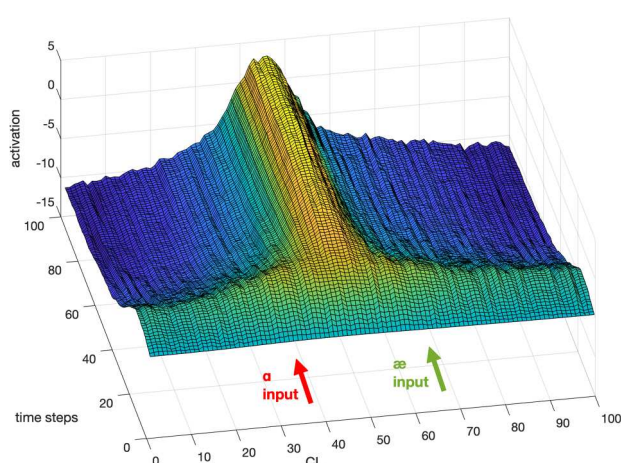


Figure 2. DFT simulation showing intermediate field stabilization for æ and ɑ inputs into a constriction location planning field.

2. Materials & methods

2.1. Participants

14 native speakers of American English participated in the experiment. All participants rated their proficiency in English as native. None of the participants reported any history of speech, language or hearing disorders. 2 participants were excluded: one based on poor understanding of the task and the second based on poor vowel formant quality in the audio recording. The results reported below are from the remaining 12 participants.

2.2. Materials

Tongue twisters consisting of vowel + /b/ segments in ABBA and BAAB patterns were constructed, based on the examples (e.g., keff geff geff keff and geff keff keff

geff) in Goldrick and Blumstein (2006). The A–B pairs were made up of the following 4 vowel pairs: /ɪ/ - /ɛ/, /æ/ - /ɑ/, /i/ - /u/, /ʊ/ - /ʌ/. Vowel pairs did not overlap in their members to ensure that if there was a trace effect in the formants, it would be in the same direction for each vowel in every trial. For example, a trace effect of /ɪ/ on an errorful production of [ɛ] would involve a lowering of F1, but a trace effect of /ɑ/ on the same vowel would mean a raising of F1. To make analyses straightforward and not have to separate trials of the same vowel based on the other pair member, each vowel only occurred in one pair. The ABBA and BAAB tongue twisters made up the critical trials. Control trials consisted of the same vowel repeated 4 times (AAAA, BBBB), as in McMillan and Corley (2010). Target sequences were presented orthographically, and participants were trained on the orthographic representation of each trial type and provided with practice trials before the main trials began. The complete set of stimuli is listed in the table below (Table 1).

A metronome sound guided participants on the production of the tongue twisters. Practice trials were said to the beat of a 60 bpm metronome, while main trials were said to the beat of a 155 bpm metronome, a little higher than Goldrick and Blumstein (2006), who used 150 bpm. Each metronome audio consisted of 4 initial beats, inserted so that participants were primed to produce the stimuli at the correct speed, followed by a beat of a higher pitch that signalled that productions should start at the next beat. There were then 20 metronome beats—leading to 5 repetitions of each trial string. The ABBA and BAAB pattern of each pair were both presented twice over the course of the experiment, giving 20 repetitions for each vowel in the critical conditions. Control trials for each vowel were repeated only once, giving, again, 20 repetitions for each vowel in a control environment. An orthographic representation of the target tongue twister remained on the screen through the duration of each trial. A fixation cross was presented between trials.

2.3. Procedure

Participants were brought into a sound attenuated booth and given a consent form, and then a questionnaire that asked for the following information: age, gender, if they speak American English and how they

would rate their proficiency in it, any other languages spoken and proficiency in each, and finally, if they had any history of language, speech, or hearing disorders. Once they had filled out the questionnaire, participants read the instructions for the tongue twister task and were told to ask the experimenter any questions—especially about the stimuli pronunciations—before they continued on to the practice trials. As mentioned previously, practice trials were set to a 60 bpm metronome. Once the practice trials were done, the instructions were reiterated, and the main trials began. Participants controlled when to start the metronome at each trial, and were told to inform the experimenter and ask questions if they were at all uncertain about the pronunciation.

2.4. Equipment

The experiment took place in a sound-attenuated booth at the Linguistics Department at Yale University. The experiment was presented using Gorilla (<https://gorilla.sc/>) (Anwyl-Irvine et al., 2020). Recordings were made via a Shure desktop microphone, and at two places—via PRAAT on the experimenter's laptop (as a continuous audio file), and a backup via Gorilla's automatic recording software for each trial.

2.5. Data analysis

All audio files were segmented using MFA forced alignment (McAuliffe et al., 2017), and this segmentation was manually corrected. Formant extraction was done via the burg method, using a Praat script written by the first author, with reference to already existing Praat formant extraction scripts (Crosswhite, 2009; DiCanio, 2013; Stanley & Lipani, 2019) (Praat: Boersma & Weenink, 2022). The formant ceiling was chosen for each participant based on visual inspection of the spectrum to ensure good formant tracking. The formant measures obtained in Praat were then hand-corrected using a Matlab script, which displays each formant estimate from PRAAT against the spectrogram (Proctor, M.). After hand correcting all formant measurements with reference to the spectrogram, we used the F1 and F2 measurements for error classification. All data and analysis scripts are available on OSF: DOI 10.17605/OSF.IO/X8AKQ.

Table 1. Experimental stimulus set.

Vowel pair	ABBA	BAAB	Control 1 (AAAA)	Control 2 (BBBB)
/æ/ - /ɑ/	ab ahb ahb ab	ahb ab ab ahb	ab ab ab ab	ahb ahb ahb ahb
/ɪ/ - /ɛ/	ib ehb ehb ib	ehb ib ib ehb	ib ib ib ib	ehb ehb ehb ehb
/i/ - /u/	eeb oob oob eeb	oob eeb eeb oob	eeb eeb eeb eeb	oob oob oob oob
/ʊ/ - /ʌ/	uhb uhb uhb uhb	uhb uhb uhb uhb	uhb uhb uhb uhb	uhb uhb uhb uhb

Below, we describe each analysis step in order. First, error classification was done using support vector machines, trained on the formant midpoints of the control productions for each participant's vowels. This formant midpoint was either F1 ("F1 vowels": /i/, /ε/, /u/, and /Λ/), or F2 ("F2 vowels": /æ/, /a/, /i/, and /u/), depending on whether differences between vowels were expected for mainly F1 or F2. This resulted in one SVM for each vowel pair (/i/ - /ε/, /æ/-/a/, /i/-/u/, /u/-/Λ/) and participant separately (SVMs: Boser et al., 1992; Burges, 1998; Cortes & Vapnik, 1995, etc.). Once trained on the control tokens, the SVMs were used to classify the critical trials as belonging to one of the vowels in each pair. The classification test and training sets were thus non-overlapping, with the training set coming from the control AAAA/BBBB trials and the test set coming from the ABBA and BAAB trials. The SVM classification was then compared to the *intention*, i.e., the segment in the ABBA/BAAB chunk each token came from. If the intention matched the classification, the token was non-errorful. If these were different, the token was labelled as "error." The SVMs were fit using the scikit learn package in python on jupyter notebook (Pedregosa et al., 2011).

Additionally, tokens with laughter or those for which a reliable formant track could not be established were removed. Together these tokens constituted 4.3% ($n = 244$) of the total number of tokens ($n = 5683$) across all participants and vowels, leaving about 96% of tokens for analysis ($n = 5439$). Additionally, we found that a sizable portion of participants, even after clarification, training and practice, were not able to produce the contrast between /u/ and /Λ/ in our stimuli. For these participants ($n = 3$), the tokens for this vowel pair were not analyzed. This left us with a total of 9 participants instead of 12 for this vowel pair. The $n = 5683$ tokens before exclusion thus came from 12 participants for 3 vowel pairs, and from 9 participants for all 4, including /u/ - /Λ/. Finally, the control tokens ($n = 1784$), which constituted the training set for the SVM, were excluded from the analysis described in the results section, giving us $n = 3655$ critical trials, out of which $n = 126$ were errorful.

The main error analysis consisted of a linear mixed effects model built to test predicted formant differences between errorful and non-errorful tokens for *both* "F2 vowels" (æ/, /a/, /i/, and /u/) and "F1 vowels" (/i/, /ε/, /u/, and /Λ/) together. In order to group all non-errors and errors under one analysis (regardless of vowel pair), we calculated a standardized formant measure based on the F2 midpoint for "F2 vowels" and the F1 midpoint for "F1 vowels." This method allows more power in the analysis, grouping the otherwise rare

error tokens together. Our method builds on an approach developed by Cohen Priva and Gleason (2020) to calculate "relative" formant values for each token. First, mean formant values (either F2 or F1) measured at the vowel midpoint were calculated for each vowel based on the control trials (of type AAAA/BBBB). Then, a log ratio (natural log) was calculated for each token, by taking the log of the ratio of each critical trial token's midpoint formant value (either F1 or F2) to the relevant category mean (i.e., $\log(F_{\text{token}}/F_{\text{CategoryMean}})$). This log ratio measure represents the deviation of each token from the vowel category mean as well as the direction of the deviation. Negative values indicate lower formants than the mean (of the control trials), while positive values indicate higher formants than the mean (of the control trials). A log ratio of zero indicates no difference from the mean. This measure enables clear comparison of critical trials (ABBA trials), which may contain coarticulation, to controls (AAAA trials), which have no vowel-to-vowel coarticulation.

In order to analyze differences between errors and non-errors across all vowels in the same analysis, we needed to ensure that the trace effect would *always* be in the same direction regardless of vowel. To achieve this, we *mirrored* half of the vowels' log ratio distributions—i.e., we multiplied their tokens' log ratios by -1. For example, errorful [i]s are predicted to have lower F2s than non-errorful [i]s, indicating the trace of the competitor, /u/. On the other hand, errorful [u]s are predicted to have higher F2s than non-errorful [u]s. To analyze both of these vowels under a single model, log ratios for vowels whose errors should show a "raising" trace were multiplied by -1, shifting the direction in which the effect should be observed. Thus the trace effect in the dataset should always result in errorful token log ratio measures that are *lower* than non-errorful token log ratio measures.

The relative formant values described above were used as the dependent variable in the following linear model, designed to test differences between errorful and non-errorful vowels (Equation (1)). F1 and F2 vowels were grouped together, since the log ratio measure is simply a relative value, whether it comes from an F1 vowel or an F2 vowel.

$$\begin{aligned} \text{Formant log ratio} &\sim \text{Vowel Classification} \\ &+ \text{Error Classification} \\ &+ (1 + \text{Error Classification} | \text{participant}) \end{aligned} \quad (1)$$

To test for duration differences, we extracted the duration of each vowel, and calculated a log ratio duration measure using the same method as described for formants. First, mean vowel duration for each vowel type

was calculated based on control tokens. This measure then was used to calculate the relative duration of each token by taking the log ratio of each token's duration over the mean duration for its vowel type. Relative duration was used as the dependent variable in the linear model in (Equation (2)) and used to test duration differences between errorful and non-errorful vowels.

$$\text{Duration log ratio} \sim \text{Vowel Classification} * \text{Error Classification} + (1 | \text{participant}) \quad (2)$$

The random effect structure was decided based on a balance between maximality and fit (Bates et al., 2015). For example, a model including a by-participant slope for duration (as was done for the formant model) did not converge, and the variance-covariance matrix for the random effects revealed variances that were almost 0. For this reason, we eliminated random slopes from the duration model. We note that the dependent measure in both models is already a normalized measure and thus will remove some participant-based variation.

3. Results

First, we report the error rates across participants for each vowel pair. Note that this rate includes participants who did not produce any errors at all for a given pair, as we want to get a sense of how often these errors were made over all participants (range across participants: 0.3%–9.8%).

Table 2 shows differences in error rates among the vowel pairs—while /ɪ/ - /ɛ/ and /i/ - /u/ tongue twisters led to similar error rates (and ones comparable to consonant studies like Goldrick & Blumstein, 2006), /æ/ - /ɑ/ and /ʊ/ - /ʌ/ tongue twisters led to higher error rates. /ʊ/ - /ʌ/ results are especially striking, with almost double the already high /æ/ - /ɑ/ error rates.

We also present the direction of errors within each pair—i.e., how many errors were produced for each vowel type (Table 3). There are some striking asymmetries—for example, intended /æ/s, /ʌ/s, and /ɛ/s led to errors more frequently than their competitors, /ɑ/s, /ʊ/s and /i/s. On the other hand, intended /i/s and /u/s led to about the same frequency of errors.

Table 2. Token counts for all vowel pairs, with error rates.

Vowel Pair	Critical Tokens (error + non-error)	Errors	Error Rate
/æ/ - /ɑ/	967	43	4.45%
/ɪ/ - /ɛ/	967	17	1.76%
/i/ - /u/	975	16	1.64%
/ʊ/ - /ʌ/	746	50	6.7%
Total	3655	126	3.45%

Table 3. Number of errors by individual vowel.

Vowel Intention	Error Category	Number of Errors
æ	ɑ	31
ɑ	æ	12
ɛ	ɪ	5
ɪ	ɛ	12
u	i	7
i	u	9
ʌ	ʊ	36
ʊ	ʌ	14

3.1. Linear model analysis

Below, we present the results from the linear models built to test error vs. non-error differences in formant values and duration, based on log ratio measures.

3.1.1. Formant analysis

The final model specification was justified via a nested model comparison using anova. The comparisons included a base model without either of the first two parenthetical terms in Equation (3), a model with the addition of the "Error Classification" term, a model with the addition of the "Vowel classification" term, and one with the addition of both. Finally, a model with their interaction, Equation (4), was also tested. This comparison revealed some interesting patterns.

$$\begin{aligned} \text{Formant log ratio} &\sim (\text{Vowel Classification}) \\ &+ (\text{Error Classification}) \\ &+ (1 + \text{Error Classification} | \text{participant}) \end{aligned} \quad (3)$$

$$\begin{aligned} \text{Formant log ratio} &\sim \text{Vowel Classification} * \text{Error Classification} \\ &+ (1 + \text{Error Classification} | \text{participant}) \end{aligned} \quad (4)$$

A comparison of the base model to the models with additional terms showed that the addition of the "Error Classification" term to a model with only "Vowel Classification" as a fixed effect significantly improved model fit ($p < 0.01$), indicating that error classification makes a significant difference to log ratio measures. The anova comparison for these models is provided in Table 4. Lastly, the addition of the interaction term did not improve the model. The fixed effects details for the final model, without the interaction term, are given in Table 5.

Table 4. Anova model comparison for formant model.

Model Structure	AIC	χ^2	df	p
Random effects only	-3565.2			
Random effects + Error Classification	-3570.0	6.74	1	< 0.01*
Random effects + Vowel Classification	-3668.8	110.81	6	< 0.01*
Random effects + Vowel Classification + Error Classification	-3673.7	6.93	1	< 0.01*
Random effects + Vowel Classification * Error Classification	-3665.4	5.70	7	0.57

Table 5. Fixed effects for formant model.

	Estimate	SE	df	<i>t</i>	<i>p</i>
(Intercept)	0.01	0.01	22.45	1.2	0.24
æ	−0.04	0.009	3640	−3.75	< 0.01*
ɪ	0.009	0.009	3637	0.98	0.33
ɛ	−0.02	0.009	3640	−2.02	< 0.05*
ʌ	−0.01	0.01	3647	−1.25	0.21
ʊ	0.06	0.01	3650	6.28	< 0.01*
i	−0.03	0.009	3636	−2.73	< 0.01*
u	−0.004	0.009	3634	−0.46	0.64
Error	−0.1	0.03	9.19	−3.46	< 0.01*

Let us now take a look at plots based on the linear model estimated marginal means (using the emmeans package; Lenth, 2024). Figure 4 shows a trace effect across vowels, i.e., a log ratio measure for the errors (Estimated Mean = −0.09, *SE* = 0.03) that is lower than that for the non-errors (Estimated Mean = 0.01, *SE* = 0.01). A comparison of estimated means using the pairs() function in emmeans revealed that this difference was statistically significant (Difference Estimate (non − error − error) = 0.103, $t(12.1) = 3.19$, $p < 0.01$). Recall that lower log ratio measures was our prediction for all vowels. Finally, we can see the estimated marginal means for errorful and non-errorful tokens by vowel (Figure 3).

The results are clear. First, we see that the log ratio measure for the non-errorful tokens is very close to 0. In fact, a test of significance for difference from 0 using the emmeans test() function revealed that the non-error estimate was not significantly different from 0 ($t(13.2) = 1.053$, $p = 0.3$), while the error estimate was

($t(12.6) = -2.672$, $p < 0.05$). This is not surprising—since the log ratio measure is a comparison to controls, this tells us that the non-errorful tokens in critical trials are not very different from their control counterparts, while the errors differ both from the controls and the critical non-errors in a systematic way. That is, the error log ratios are lower than both non-errorful tokens and controls, indicating formant values that are shifted towards the competitor vowel.

We also note that the estimated marginal means plots are qualitatively similar to the raw plots of the dependent variable (Figures A3–A4), which can be found in the Appendix.

3.1.2. Duration analysis

The final duration model was arrived at using the same method as used for formants.

$$\begin{aligned} \text{Duration log ratio} \sim & (\text{Vowel Classification}) \\ & + (\text{Error Classification}) \\ & + (1 + \text{Error Classification} \mid \text{participant}) \end{aligned} \quad (5)$$

$$\begin{aligned} \text{Duration log ratio} \sim & \text{Vowel Classification} * \text{Error} \\ & \text{Classification} \\ & + (1 + \text{Error Classification} \mid \text{participant}) \end{aligned} \quad (6)$$

For the duration models, the pattern was strikingly different. While the addition of the “Vowel Classification” term improved the model, the addition of “Error Classification” did not—both when it was added as the sole

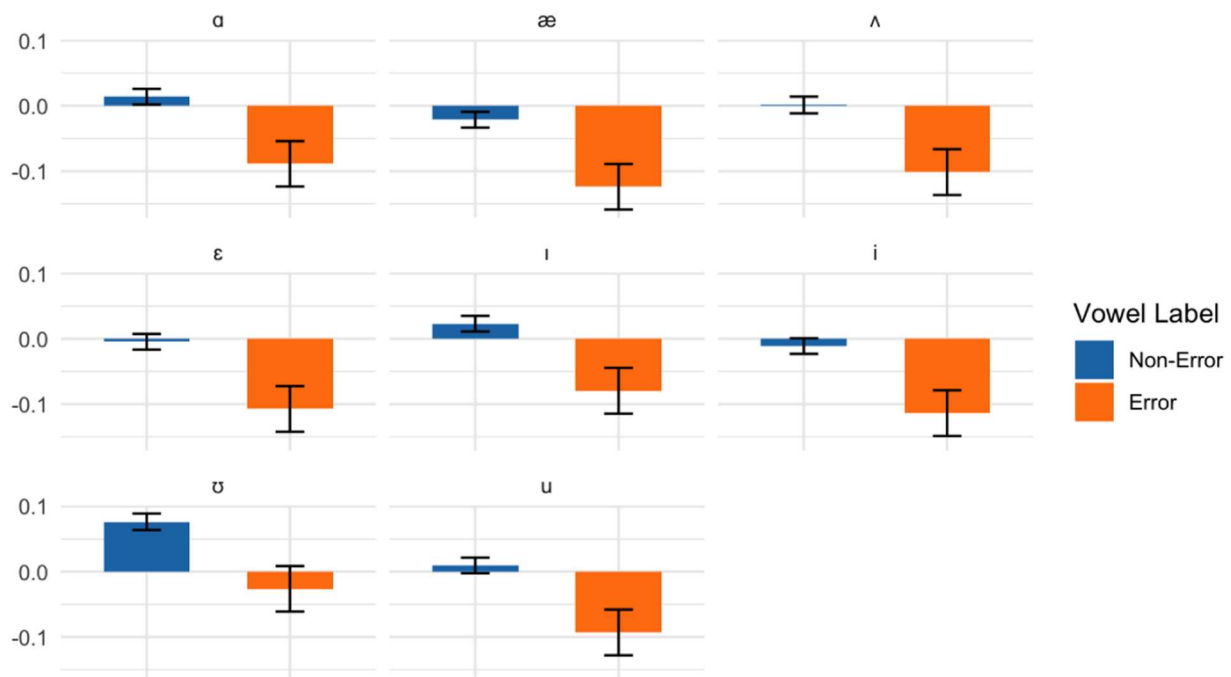


Figure 3. Estimated formant log ratio means for errors and non-errors by vowel. Error bars represent 1 standard error.

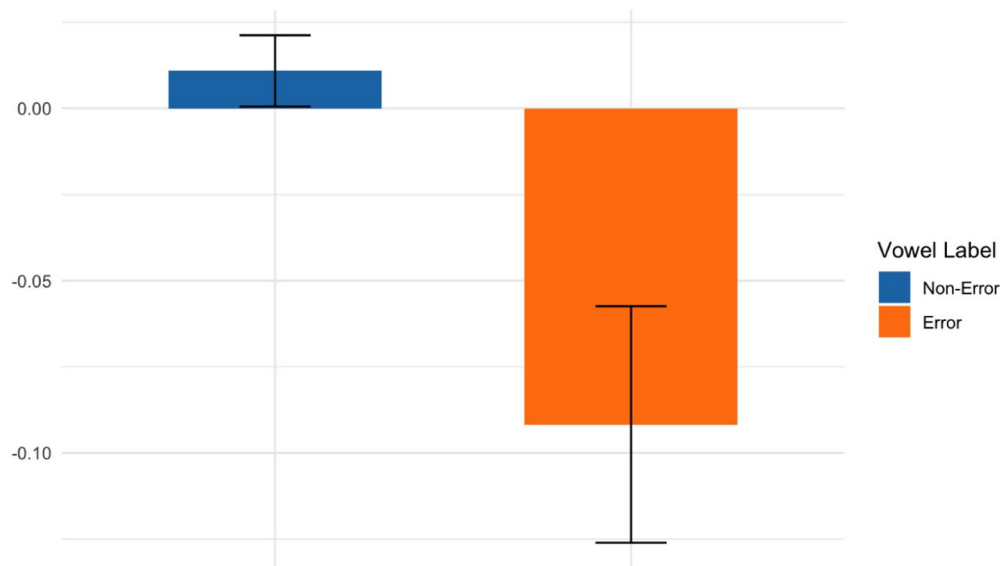


Figure 4. Estimated formant log ratio means for errors and non-errors across vowels. Error bars represent 1 standard error.

term to the base model, and when it was added after “Vowel Classification.” However, the best model was the one which included an *interaction* between “Vowel Classification” and “Error Classification.” The anova comparison is presented in Table 6. An investigation of the estimated marginal means plot (Figure 5) shows us why—errorful token log ratios differ across vowels. They are sometimes longer than their non-errorful counterparts, and sometimes shorter. This stands in stark contrast to the formant analysis, which showed consistent differences between errors and non-errors across vowels. Figure 6 indicates that neither non-errors nor errors are different from 0, and this is confirmed using the emmeans test() function ($p > 0.5$ for both). The fixed effects details for the model with an interaction term are provided in Table 7.

Plots of the raw duration data can be found in the Appendix (Figures A5–A6).

3.2. Summary of results

In our study, we found vowel errors at rates comparable to previous consonant studies. There were some asymmetries among the vowel pairs, with half of the vowels having more errorful counterparts. Linear

model analysis revealed differences in errorful and non-errorful vowel formants, and no differences for duration. We thus conclude that vowel errors show a trace effect in the dimension of formant measures, as predicted by the Stern et al. (2022).

3.3. Online correction

In addition to the quantitative analysis described above, we also noticed some cases in which vowel quality changes over the course of the vowel interval. We refer to these tokens as online corrections, or “OC’s.” OC’s start as categorical errors but show an increasing influence of the intended vowel over time. They were heard as diphthongs made up of first the error, and then the intended vowel. There were a total of 15 OCs in the data. Out of the four vowel pairs, i-u led to the most OCs (with 9) followed by æ-ɑ (with 4), and i-ε (with just two). Interestingly, ʊ-ʌ did not lead to any (visually or auditorily perceptible) OCs. This parenthetical is an important methodological point. Out of the 15 OCs, only 2 were classified as errors using our SVM method, which was based on the midpoint of the vowel formants. If this classification had been done perceptually, perhaps a different decision would have been made. On the other hand, the existing, machine-based, classification may be more accurate considering these OCs show correction across the trajectory (i.e., they become more like the intention, and less errorful, over time).

Two examples of these OCs, for the vowel pair i - u, are given below (Figure 7). In the first, the token is incorrectly produced as an [u], and corrected to the intention, /i/ over

Table 6. Anova model comparison for duration model.

Model Structure	AIC	χ^2	df	p
Random effects only	−2203.5			
Random effects + Error Classification	−2202.3	0.84	1	0.36
Random effects + Vowel Classification	−2251.4	61.04	6	< 0.01*
Random effects + Vowel Classification + Error Classification	−2249.8	0.39	1	0.53
Random effects + Vowel Classification * Error Classification	−2273.6	37.8	7	< 0.01*

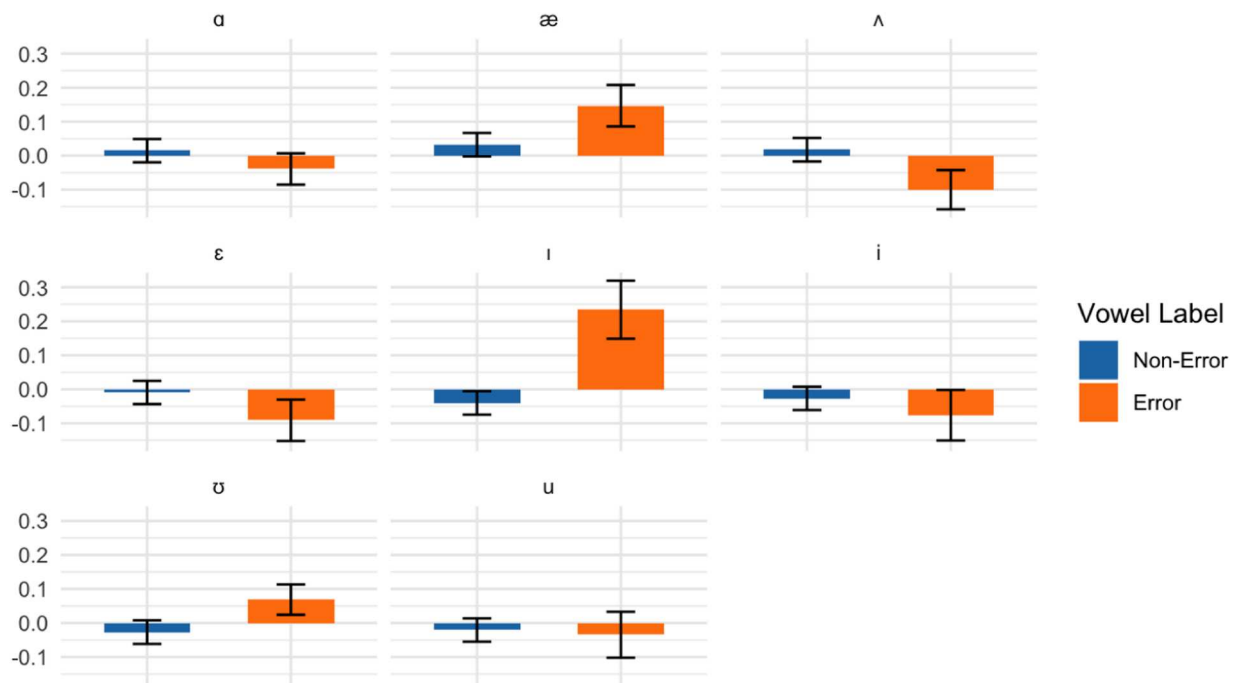


Figure 5. Estimated duration log ratio means by vowel. Error bars represent 1 standard error.

time. The second shows the opposite pattern—an incorrectly produced [i] is corrected to the intention, /u/. This can be seen in the movement of the F2 trajectory over time.

The OCs highlight the benefit of looking at the acoustics of vowels as compared to consonants. Vowels are typically longer in duration than consonants. This affords the opportunity to observe how articulation and the resulting acoustics evolve over a longer temporal interval. This dynamicity is what allows us to note correction over time in the first place. Further, these OCs may

have been an observation of Laver (1980)'s vowel errors study as well, which noted diphthongal errors combining two target vowels.

OC may be predicted by both models we introduced above. On Goldrick and Chu (2014)'s gesture-based account, with a reduced errorful gesture and the intention present at the same time, we might see one gesture terminate just as the other gesture target takes over the tongue body, the relevant articulator for vowels. This could give rise to the formant transitions

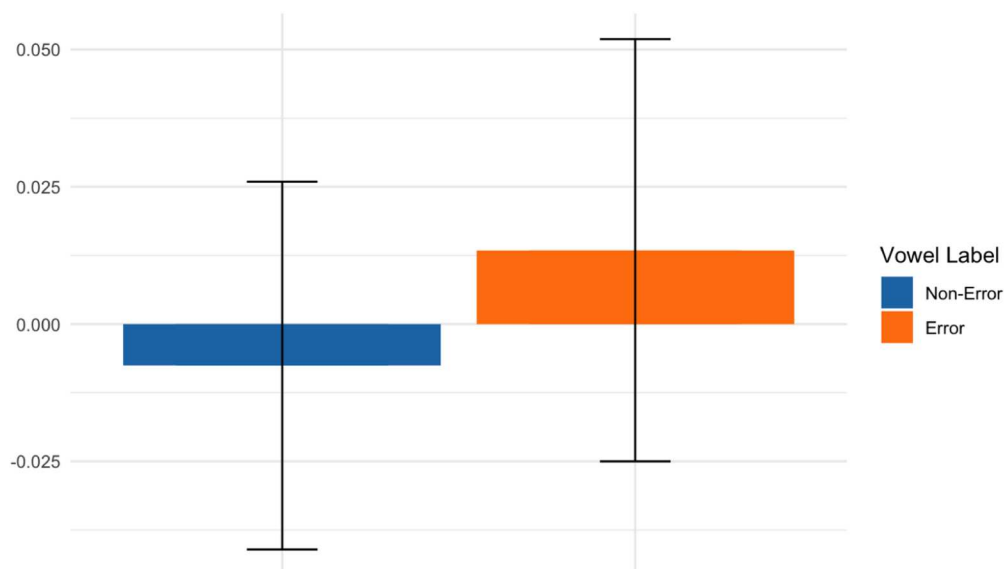


Figure 6. Estimated duration log ratio means across vowels. Error bars represent 1 standard error.

Table 7. Fixed effects for duration model.

	Estimate	SE	df	<i>t</i>	<i>p</i>
(Intercept)	0.01	0.03	13.33	0.44	0.67
æ	0.02	0.01	3643	1.53	0.13
ɪ	−0.05	0.01	3643	−4.79	< 0.01*
ɛ	−0.02	0.01	3643	−2.1	< 0.05*
ʌ	0.003	0.01	3645	0.23	0.82
ʊ	−0.04	0.01	3645	−3.35	< 0.01*
i	−0.04	0.01	3643	−3.67	< 0.01*
u	−0.04	0.01	3643	−3.10	< 0.01*
Error	−0.05	0.03	3644	−1.65	0.10
æ * Error	0.17	0.06	3644	2.76	< 0.01*
ɪ * Error	0.33	0.09	3643	3.85	< 0.01*
ɛ * Error	−0.03	0.06	3643	−0.46	0.64
ʌ * Error	−0.06	0.06	3643	−1.11	0.27
ʊ * Error	0.15	0.04	3644	3.34	< 0.01*
i * Error	0.005	0.07	3643	0.06	0.95
u * Error	0.04	0.07	3643	0.59	0.55

in our OC tokens. In DFT, transition from one vowel target to the other would have to involve multiple activation peaks. An activation peak forms first for the error, subsequently falling below activation threshold while the intention rises above threshold, causing a second activation peak and the transition to non-errorful formant values. Such transitions are predicted to occur naturally in non-errorful speech as well, leading to observed patterns in pitch (Chaturvedi & Shaw, 2025) and voicing (Shaw, 2025).

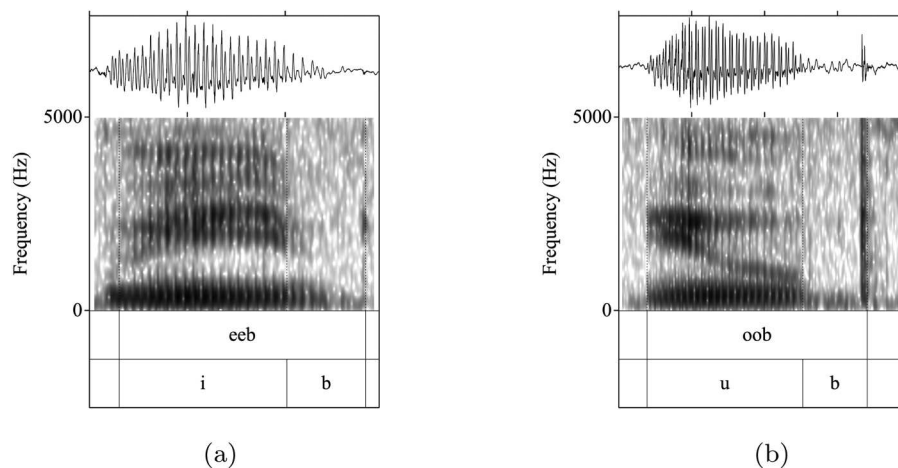
Finally, we also saw a pattern opposite to online correction, showing a change to the “incorrect” vowel over time ($n=2$ for the pair /æ/-/a/). These tokens cover another possibility that is predicted by both models—i.e., a move towards the wrong vowel target due to the activation of an *incorrect intention*. Given the nature of the task—fast repetition of an alternating sequence—it is possible that speakers lose track of the specific segment of the tongue twister, leading to “correction” in the wrong direction. In the GSP model, this

could happen in the same manner as for OCs, but with the “correct” gesture terminating during vowel articulation to give way to the incorrect gesture. In the DNF model, it is possible that the incorrect vowel representation receives a boost during articulation, allowing the stabilization of an activation peak at an “incorrect” formant value.

4. Discussion

The phonetic details of speech provide a rich source of information about the activation, competition and selection of linguistic representations in language production. Past work has shown that pronunciations perceived as typical (non-errorful) may include an intrusion of a competitor phonological form (Goldstein et al., 2007; McMillan & Corley, 2010; Pouplier, 2007), detectable in the phonetic details of speech. Additionally, speech errors in consonants show a phonetic trace of the intended sound (Alderete et al., 2021; Goldrick & Blumstein, 2006). Thus, speech from non-clinical, neuro-typical populations involves the phonetic influence of multiple phonological representations, regardless of whether it is judged to be errorful or non-errorful. In this study, we added to this literature by investigating vowel errors using a similar experimental paradigm used to investigate consonants in past work.

The results confirm, firstly, that vowel errors can be experimentally elicited in the same manner as consonants. In fact, error rates, while variable across vowel pairs, are comparable to or higher than consonant error rates. This finding is non-trivial because it diverges from previous studies of vowel errors in which errors were determined by listener perception (Rusaw & Cole, 2011; Shattuck-Hufnagel, 1986), instead of classification

**Figure 7.** F2 Trajectories for blends in the vowel pair i - u. (a) Intention: /i/ and (b) Intention: /u/.

through machine learning as in the current study. There are many possible reasons for this divergence. For starters, the tasks were different across studies, so it could be that the frequency of vowel errors interacts with task. Alternatively, it could be the method of determining errors. Perhaps listeners would not detect the same tokens as errors as our machine learning method. There is some evidence that vowels constrain lexical access to a lesser degree than consonants, i.e., a change in vowel has less impact on lexical access than a change in consonant (Cutler et al., 2000; Van Ooijen, 1996). If listeners tolerate more variation in vowels than in consonants, they may be less likely to recognize a vowel error in a real word. This may be true for English in part because naturally occurring accent variation tends to involve more differences in vowels than in consonants (Wells, 1982). When it comes to categorization tasks, English listeners are known to tolerate substantial variation in vowel formants in non-words as well, at least in naturally produced vowels from different accents (Shaw et al., 2018). Determining the source of reported differences in vowel error rates across past studies, including possible differences in word and non-word items, is an important direction for future research.

The results of our experiments also showed error rate differences *across* vowels. The most striking was the higher number of errors for the following error directions: (1) /æ/ → [a], and (2) /ʌ/ → [u]. We are not sure why these vowel pairs (and directions) were particularly prone to errors. It is possible that the written form of the vowels in our experiment contributed in part to higher numbers of errors on these vowels. Both vowel pairs showing particularly high error rates had overlapping orthography in the stimuli: “a” was used for both /æ/ - /a/ (e.g., ab ahb ahb ab), and “u” was used for both /u/ - /ʌ/ (e.g., ub uhb uhb ub) (Table 1). The other two vowel pairs, with comparable number of errors across both directions, did not share stimuli orthography. Interestingly, if shared orthography is the reason for higher numbers of errors, the error rates could potentially be even *more* like consonant error rates rather than slightly higher. That is, the consonants in past studies, which did not share an orthographic representation, have similar error rates to the vowels in our study that also did not share an orthographic representation. Nevertheless, shared orthography cannot account for the asymmetry in error directions for both pairs, with the large number of errors coming only in the /æ/ → [a] and /ʌ/ → [u] directions.

Our linear model of F1 and F2 log ratio values revealed significant trace effects in the expected direction. Moreover, the results rule out the possibility that the observed effects on errors were due to vowel-to-

vowel coarticulation instead of “competitor” effects in speech planning. The critical trials were produced with alternating ABBA stimuli. Because of this, they may have had coarticulatory influences, i.e., vowel-to-vowel coarticulation. The direction of such a coarticulatory influence would presumably be assimilatory, which is the same direction of “competitor” influence in the trace effect. However, our analysis allows us to rule out coarticulation as a possible account of phonetic patterns in errorful vowels. This is because any vowel-to-vowel coarticulatory influence in the critical trials would be present on *all* vowels, whether errorful or not. Following an innovation of McMillan and Corley (2010)’s design, our experiment included control trials, e.g., AAAA, with no alternation between vowels. Our key dependent variable, the log ratio measure, rendered tokens in the alternating critical trials relative to the mean of the non-alternating controls. We found that the log ratio of the non-errors was not significantly different from zero, indicating that any coarticulatory effects owing to the alternating nature of the stimuli in critical trials were negligible, at least as measured at the vowel midpoint. On the other hand, errorful log ratios were significantly different from zero. The source of this difference cannot be coarticulation since the effect was absent on non-errors. We therefore conclude that the trace effect observed in errors has its source in speech planning as opposed to coarticulation.

The results of this study provide further evidence for “cascading activation” in speech production (Goldrick & Blumstein, 2006). Vowel errors, like consonants, show traces of the intended sound in the phonetics, indicating that both the intended and errorful sound representations are active at some level of language production. Selection at the level of the phoneme is not strictly categorical, a result which converges with a host of other results showing effects of lexical competition on phonetics (see introduction).

The phonetics of vowel errors helped us to adjudicate between two models of speech production planning that we introduced in Section 1.2. Goldrick and Chu (2014)’s model—which specifies shortened gesture durations for speech errors—predicts that errorful vowels will be shorter than non-errorful vowels. Our results did not support this prediction. On the other hand, Stern et al. (2022)’s model predicts trace effects in dimensions of articulatory control, a prediction that is supported by formant measurements. This means that, at the very least, the GSC and gesture duration model (Goldrick & Chu, 2014) needs some modification in the way that trace effects are derived. Pouplier and Goldstein (2014), for example, suggest that gesture activation may be a better parameter (than duration) to link gradient

symbolic activations in GSC to gestural dynamics. Since both models rely crucially on cascading activation, it is possible as well that they can be integrated. GSC may provide input activation values to DNFs (representing activations of two phonemic representations) that determine phonetic values on the basis of gradient activation.

Advancing language production models is an important priority for integrating results across tasks and populations, including comparisons between typical and atypical speech populations. Our results indicate that regardless of established processing and brain region differences (e.g., Boatman et al., 1997; Caramazza et al., 2000), vowels and consonants may have similar *mechanisms* supporting their productions, just based on different dimensions of articulation. Errors in both vowels and consonants exhibit trace effects that are consistent with multiple inputs into a DNF. In contrast, non-errorful speech often involves inhibitory inputs to the relevant planning fields (Chaturvedi & Shaw, 2025; Stern & Shaw, 2023). We can therefore understand speech errors elicited in experimental tasks from neuro-typical, non-clinical populations as a failure to inhibit a competitor representation under the conditions of the task.

Neural control of inhibition is an important dimension of variation across clinical populations. Parkinson's Disease (PD) patients, for example, exhibit task-switching difficulties (e.g., Benecke et al., 1987), hypothesized to arise partly from a failure to inhibit competing actions (Inzelberg et al., 2001; Obeso et al., 2011; Rubchinsky et al., 2003; Spencer & Rogers, 2005). This failure to inhibit is also hypothesized to underlie word-processing deficits in PD, pointing to a more general inhibition deficit (Marí-Beffa et al., 2005), also apparent in speech behaviour (Thies et al., 2020). Thies et al. (2020) showed that PD patients who scored lower on task-switching, based on a general metric (the trail-making task), produced longer and more extreme f_0 values on accented syllables than a control group, suggesting a difficulty for PD patients in transitioning away from a pitch target. Additionally, PD patients with lower motor ability (based on UPDRS II score) had reduced vowel spaces. Both of these correlations could be related to degraded ability of PD patients to inhibit competitor representations. Inhibiting a DNF upon achievement of a speech target facilitates transition to the next speech target (Chaturvedi & Shaw, 2025) and can lead to more extreme speech targets, i.e., hyper-articulation (Stern & Shaw, 2023) in neuro-typical, non-clinical populations. We effectively model the trace effect in speech errors by eschewing competitor inhibition, allowing multiple inputs to shape evolution of a speech planning field. This may

be the standard state of affairs for PD patients and other clinical populations who have general difficulties with neural inhibition.

Further, the presence of OCs in our dataset, while rare, provide evidence for online correction mechanisms like those suggested by auditory perturbation studies (e.g., Houde & Jordan, 1998; Mitsuya et al., 2011; Niziolek et al., 2013). The “efference copy,” a prediction (generated by the speaker) that is compared to auditory feedback, has been hypothesized to be the basis of such correction in normal populations (e.g., Hickok, 2012) and important in first language acquisition (Byun et al., 2016). Recent research using perturbed auditory feedback paradigms in aphasia populations has shown that prediction-feedback comparison mechanisms may be impaired in patients with lesions to left hemisphere sensorimotor networks (Behroozmand et al., 2022). The presence of OCs shows that online correction mechanisms are at play not only when there is a mismatch between motor movements and their acoustic consequences (as in perturbation studies), but also when there is a mismatch between higher level intentions and motor commands/acoustics (speech errors). This supports the conclusion that a breakdown in normal feedback systems plays a role in speech-related deficits in Aphasia (where there is no perturbed feedback). In both of these cases, modelling speech errors unifies observations of language production in clinical populations with those of neuro-typical populations under the specific conditions of an experimental task. This approach illustrates how theoretically-driven experimentation and clinical research can jointly inform the cognitive neuropsychology of language.

5. Conclusion

We generalized a paradigm for eliciting speech errors in consonants to vowels, finding that experimentally-induced vowel errors are at least as common as consonant errors and that vowel errors show phonetic traces of the intended segment (“trace effects”), just as consonant errors have been shown to do. The vowel data further allowed us to adjudicate between competing models of how phonological competition influences the phonetic signal. Our data provide evidence for the proposal that trace effects derive from a failure to inhibit competitors. In this way, the mechanism underlying experimentally-induced speech errors relates directly to neural accounts of pathological speech in clinical populations, such as Parkinson's disease, where a general deficit of inhibitory control impacts speech and non-speech motor behaviour alike.

Acknowledgments

Thank you to the editor, Svetlana Pinet, and the anonymous reviewers, whose feedback and suggestions greatly improved the manuscript. We would also like to thank the members of the Phonology & Phonetics reading group, and the Language & Brain Lab at Yale (with special thanks to Michael Stern, Alessandra Pintado-Urbanc, and Teresa Borneo). Thank you to the participants at and reviewers for SYNC 2023, the participants at the PhPhoNE conference 2024, especially Joe Pater, and the participants at IWOLP 2024 for their comments and suggestions. We are grateful to Doug Whalen for his guidance and comments, and Michael Proctor for providing us with his formant correction script. All errors are our own.

Data availability statement

The data supporting the results of this study are available from OSF with DOI 10.17605/OSF.IO/X8AKQ.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

ORCID

Manasvi Chaturvedi  <http://orcid.org/0009-0001-5892-9897>

References

- Alderete, J., Baese-Berk, M., Leung, K., & Goldrick, M. (2021). Cascading activation in phonological planning and articulation: Evidence from spontaneous speech errors. *Cognition*, 210, 104577. <https://doi.org/10.1016/j.cognition.2020.104577>
- Anwyl-Irvine, A. L., Massonnié, J., Flitton, A., Kirkham, N., & Evershed, J. K. (2020). Gorilla in our midst: An online behavioral experiment builder. *Behavior Research Methods*, 52(1), 388–407. <https://doi.org/10.3758/s13428-019-01237-x>
- Baese-Berk, M., & Goldrick, M. (2009). Mechanisms of interaction in speech production. *Language and Cognitive Processes*, 24(4), 527–554. <https://doi.org/10.1080/01690960802299378>
- Bates, D., Kliegl, R., Vasishth, S., & Baayen, H. (2015). Parsimonious mixed models. *arXiv preprint arXiv:1506.04967*.
- Beach, S. D., & Niziolek, C. A. (2024). Inhibitory modulation of speech trajectories: Evidence from a vowel-modified stroop task. *Cognitive Neuropsychology*, 41(1–2), 51–69. <https://doi.org/10.1080/02643294.2024.2315831>
- Behroozmand, R., Bonilha, L., Rorden, C., Hickok, G., & Fridriksson, J. (2022). Neural correlates of impaired vocal feedback control in post-stroke aphasia. *Neuroimage*, 250, 118938. <https://doi.org/10.1016/j.neuroimage.2022.118938>
- Benecke, R., Rothwell, J., Dick, J., Day, B., & Marsden, C. (1987). Disturbance of sequential movements in patients with Parkinson's disease. *Brain*, 110(2), 361–379. <https://doi.org/10.1093/brain/110.2.361>
- Blackwood Ximenes, A., J. A. Shaw, & Carignan, C. (2017). A comparison of acoustic and articulatory methods for analyzing vowel differences across dialects: Data from American and Australian English. *The Journal of the Acoustical Society of America*, 142(1), 363–377. <https://doi.org/10.1121/1.4991346>
- Boatman, D., Hall, C., Goldstein, M. H., Lesser, R., & Gordon, B. (1997). Neuroperceptual differences in consonant and vowel discrimination: As revealed by direct cortical electrical interference. *Cortex*, 33(1), 83–98. [https://doi.org/10.1016/S0010-9452\(97\)80006-8](https://doi.org/10.1016/S0010-9452(97)80006-8)
- Boersma, P., & Weenink, D. (2022). *Praat: Doing phonetics by computer* [computer program].
- Boser, B. E., Guyon, I. M., & Vapnik, V. N. (1992). A training algorithm for optimal margin classifiers. In *Proceedings of the Fifth Annual Workshop on Computational Learning Theory, COLT '92, New York, NY, USA* (pp. 144–152). Association for Computing Machinery.
- Browman, C. (1990). Tiers in articulatory phonology, with some implications for casual speech. In *Laboratory phonology I: Between the grammar and the physics of speech*. Cambridge University Press.
- Burges, C. J. (1998). A tutorial on support vector machines for pattern recognition. *Data Mining and Knowledge Discovery*, 2(2), 121–167. <https://doi.org/10.1023/A:1009715923555>
- Byun, T. M., Inkelas, S., & Rose, Y. (2016). The a-map model: Articulatory reliability in child-specific phonology. *Language*, 92(1), 141–178. <https://doi.org/10.1353/lan.2016.0000>
- Caramazza, A., Chialant, D., Capasso, R., & Miceli, G. (2000). Separable processing of consonants and vowels. *Nature*, 403(6768), 428–430. <https://doi.org/10.1038/35000206>
- Chaturvedi, M., & Shaw, J. A. (2025). A dynamic neural model of tonal downstep. In *Proceedings of the Annual Meetings on Phonology* (Vol. 1). University of Massachusetts Amherst Libraries.
- Chodroff, E., & Wilson, C. (2017). Structure in talker-specific phonetic realization: Covariation of stop consonant VOT in American English. *Journal of Phonetics*, 61, 30–47. <https://doi.org/10.1016/j.wocn.2017.01.001>
- Cohen Priva, U., & Gleason, E. (2020). The causal structure of lenition: A case for the causal precedence of durational shortening. *Language*, 96(2), 413–448. <https://doi.org/10.1353/lan.2020.0025>
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297. <https://doi.org/10.1023/A:1022627411411>
- Crosswhite, K. (2009). Formant logger. Accessed from <https://queenoflily.blogspot.com/2008/11/praat-script-resources.html>.
- Cutler, A., Sebastián-Gallés, N., Soler-Vilageliu, O., & B. Van Ooijen (2000). Constraints of vowels and consonants on lexical selection: Cross-linguistic comparisons. *Memory & Cognition*, 28(5), 746–755. <https://doi.org/10.3758/BF03198409>
- Dell, G. S. (1986). A spreading-activation theory of retrieval in sentence production. *Psychological Review*, 93(3), 283–321. <https://doi.org/10.1037/0033-295X.93.3.283>
- Dell, G. S., Reed, K. D., Adams, D. R., & Meyer, A. S. (2000). Speech errors, phonotactic constraints, and implicit learning: A study of the role of experience in language

- production. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 26(6), 1355–1367.
- DiCanio, C. (2013). Formant script for praat (time normalized). <https://www.acsu.buffalo.edu/cdicanio/scripts.html>.
- Eimas, P. D. (1963). The relation between identification and discrimination along speech and non-speech continua. *Language and Speech*, 6(4), 206–217. <https://doi.org/10.1177/002383096300600403>
- Frisch, S. A., & Wright, R. (2002). The phonetics of phonological speech errors: An acoustic analysis of slips of the tongue. *Journal of Phonetics*, 30(2), 139–162. <https://doi.org/10.1006/jpho.2002.0176>
- Fromkin, V. A. (1971). The non-anomalous nature of anomalous utterances. *Language*, 47(1), 27–52. <https://doi.org/10.2307/412187>
- Fry, D. B., Abramson, A. S., Eimas, P. D., & Liberman, A. M. (1962). The identification and discrimination of synthetic vowels. *Language and Speech*, 5(4), 171–189. <https://doi.org/10.1177/002383096200500401>
- Garrett, M. F. (1975). The analysis of sentence production. In *Psychology of learning and motivation* (Vol. 9, pp. 133–177). Elsevier.
- Goldrick, M. (2006). Limited interaction in speech production: Chronometric, speech error, and neuropsychological evidence. *Language and Cognitive Processes*, 21(7-8), 817–855. <https://doi.org/10.1080/01690960600824112>
- Goldrick, M., & Blumstein, S. E. (2006). Cascading activation from phonological planning to articulatory processes: Evidence from tongue twisters. *Language and Cognitive Processes*, 21(6), 649–683. <https://doi.org/10.1080/01690960500181332>
- Goldrick, M., & Chu, K. (2014). Gradient co-activation and speech error articulation: Comment on Pouplier and Goldstein (2010). *Language, Cognition and Neuroscience*, 29(4), 452–458. <https://doi.org/10.1080/01690965.2013.807347>
- Goldrick, M., Keshet, J., Gustafson, E., Heller, J., & Needle, J. (2016). Automatic analysis of slips of the tongue: Insights into the cognitive architecture of speech production. *Cognition*, 149, 31–39. <https://doi.org/10.1016/j.cognition.2016.01.002>
- Goldrick, M., Ross Baker, H., Murphy, A., & Baese-Berk, M. (2011). Interaction and representational integration: Evidence from speech errors. *Cognition*, 121(1), 58–72. <https://doi.org/10.1016/j.cognition.2011.05.006>
- Goldstein, L., Pouplier, M., Chen, L., Saltzman, E., & Byrd, D. (2007). Dynamic action units slip in speech production errors. *Cognition*, 103(3), 386–412. <https://doi.org/10.1016/j.cognition.2006.05.010>
- Hickok, G. (2012). Computational neuroanatomy of speech production. *Nature Reviews Neuroscience*, 13(2), 135–145. <https://doi.org/10.1038/nrn3158>
- Houde, J. F., & Jordan, M. I. (1998). Sensorimotor adaptation in speech production. *Science*, 279(5354), 1213–1216. <https://doi.org/10.1126/science.279.5354.1213>
- Inzelberg, R., Plotnik, M., Flash, T., Schechtman, E., Shahar, I., & Korczyn, A. D. (2001). Mental and motor switching in parkinson's disease. *Journal of Motor Behavior*, 33(4), 377–385. <https://doi.org/10.1080/00222890109601921>
- Laver, J. (1980). Slips of the tongue as neuromuscular evidence for a model of speech production. In H.-W. Dechert, & M. Raupach (Eds.), *Temporal variables in speech: Studies in honour of Frieda Goldman-Eisler* (pp. 21–26). Mouton The Hague.
- Lenth, R. V. (2024). *emmeans: Estimated marginal means, aka least-squares means*. R package version 1.10.6. <https://rvinth.github.io/emmeans/>.
- Levelt, W. J. (1992). Accessing words in speech production: Stages, processes and representations. *Cognition*, 42(1–3), 1–22. [https://doi.org/10.1016/0010-0277\(92\)90038-J](https://doi.org/10.1016/0010-0277(92)90038-J)
- Levelt, W. J. (1999). Models of word production. *Trends in Cognitive Sciences*, 3(6), 223–232. [https://doi.org/10.1016/S1364-6613\(99\)01319-4](https://doi.org/10.1016/S1364-6613(99)01319-4)
- Lindblom, B. (1963). Spectrographic study of vowel reduction. *The Journal of the Acoustical Society of America*, 35(11), 1773–1781. <https://doi.org/10.1121/1.1918816>
- Lisker, L., & Abramson, A. S. (1964). A cross-language study of voicing in initial stops: Acoustical measurements. *Word*, 20(3), 384–422. <https://doi.org/10.1080/00437956.1964.11659830>
- Magen, H. S. (1997). The extent of vowel-to-vowel coarticulation in English. *Journal of Phonetics*, 25(2), 187–205. <https://doi.org/10.1006/jpho.1996.0041>
- Marí-Beffa, P., Hayes, A. E., Machado, L., & Hindle, J. V. (2005). Lack of inhibition in Parkinson's disease: Evidence from a lexical decision task. *Neuropsychologia*, 43(4), 638–646. <https://doi.org/10.1016/j.neuropsychologia.2004.07.006>
- McAuliffe, M., Socolof, M., Mihuc, S., Wagner, M., & Sonderegger, M. (2017). Montreal Forced Aligner: Trainable text-speech alignment using Kaldi. In *Proc. Interspeech 2017* (pp. 498–502). https://montreal-forced-aligner.readthedocs.io/en/v3.3.0/user_guide/index.html#citation.
- McMillan, C. T., & Corley, M. (2010). Cascading influences on the production of speech: Evidence from articulation. *Cognition*, 117(3), 243–260. <https://doi.org/10.1016/j.cognition.2010.08.019>
- Meringer, R., & Mayer, C. (1895). Versprechen und verlesen: Eine psychologisch-linguistische studie. <https://www.mpi.nl/publications/item2359139/versprechen-und-verlesen-eine-psychologisch-linguistische-studie>.
- Mitsuya, T., MacDonald, E. N., Purcell, D. W., & Munhall, K. G. (2011). A cross-language study of compensation in response to real-time formant perturbation. *The Journal of the Acoustical Society of America*, 130(5), 2978–2986. <https://doi.org/10.1121/1.3643826>
- Moon, S.-J., & Lindblom, B. (1994). Interaction between duration, context, and speaking style in english stressed vowels. *The Journal of the Acoustical Society of America*, 96(1), 40–55. <https://doi.org/10.1121/1.410492>
- Mowrey, R. A., & MacKay, I. R. A. (1990). Phonological primitives: Electromyographic speech error evidence. *The Journal of the Acoustical Society of America*, 88(3), 1299–1312. <https://doi.org/10.1121/1.399706>
- Niziolek, C. A., Nagarajan, S. S., & Houde, J. F. (2013). What does motor efference copy represent? Evidence from speech production. *Journal of Neuroscience*, 33(41), 16110–16116. <https://doi.org/10.1523/JNEUROSCI.2137-13.2013>
- Obeso, I., Wilkinson, L., Casabona, E., Bringas, M. L., Alvarez, M., Alvarez, L., Pavón, N., Rodríguez-Oroz, M.-C., Macías, R., Obeso, J. A., & Jahanshahi, M. (2011). Deficits in inhibitory control and conflict resolution on cognitive and motor tasks in Parkinson's disease. *Experimental Brain Research*, 212(3), 371–384. <https://doi.org/10.1007/s00221-011-2736-6>

- Öhman, S. E. G. (1966). Coarticulation in VCV utterances: Spectrographic measurements. *The Journal of the Acoustical Society of America*, 39(1), 151–168. <https://doi.org/10.1121/1.1909864>
- Parrell, B., & Narayanan, S. (2018). Explaining coronal reduction: Prosodic structure and articulatory posture. *Phonetica*, 75(2), 151–181. <https://doi.org/10.1159/000481099>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Peterson, R. R., & Savoy, P. (1998). Lexical selection and phonological encoding during language production: Evidence for cascaded processing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 24(3), 539–557.
- Poupplier, M. (2007). Tongue kinematics during utterances elicited with the SLIP technique. *Language and Speech*, 50(3), 311–341. <https://doi.org/10.1177/00238309070500030201>
- Poupplier, M., & Goldstein, L. (2014). The relationship between planning and execution is more than duration: Response to Goldrick & Chu. *Language, Cognition and Neuroscience*, 29(9), 1097–1099. <https://doi.org/10.1080/01690965.2013.834063>
- Rubchinsky, L. L., Kopell, N., & Sigvardt, K. A. (2003). Modeling facilitation and inhibition of competing motor programs in basal ganglia subthalamic nucleus–pallidal circuits. *Proceedings of the National Academy of Sciences*, 100(24), 14427–14432. <https://doi.org/10.1073/pnas.2036283100>
- Rusaw, E., & Cole, J. (2011). Speech error evidence on the role of the vowel in syllable structure. In *Proceedings of the International Congress on Phonetic Sciences, ICPHS XVII* (pp. 1734–1737).
- Schöner, G., & Spencer, J. P. (2016). *Dynamic thinking: A primer on dynamic field theory*. Oxford University Press.
- Shattuck-Hufnagel, S. (1986). The representation of phonological information during speech production planning: Evidence from vowel errors in spontaneous speech. *Phonology*, 3, 117–149. <https://doi.org/10.1017/S0952675700000609>
- Shaw, J., Best, C. T., Docherty, G., Evans, B. G., Foulkes, P., Hay, J., & Mulak, K. E. (2018). Resilience of english vowel perception across regional accent variation. *Laboratory Phonology*, 9(1), 1–36. <https://doi.org/10.5334/labphon.87>
- Shaw, J. A. (2025). Unifying phonological and phonetic aspects of speech in dynamic neural fields: The case of laryngeal patterns in Japanese. *Phonological Studies*, 28, 1–12.
- Smolensky, P., Goldrick, M., & Mathis, D. (2014). Optimization and quantization in gradient symbol systems: A framework for integrating the continuous and the discrete in cognition. *Cognitive Science*, 38(6), 1102–1138. <https://doi.org/10.1111/cogs.2014.38.issue-6>
- Spencer, K. A., & Rogers, M. A. (2005). Speech motor programming in hypokinetic and ataxic dysarthria. *Brain and Language*, 94(3), 347–366. <https://doi.org/10.1016/j.bandl.2005.01.008>
- Stanley, J., & Lipani, L. (2019). Automatic formant extraction in praat. Retrieved from: https://joeystanley.com/downloads/191002-formant_extraction
- Stern, M. C., Chaturvedi, M., & Shaw, J. A. (2022). A dynamic neural field model of phonetic trace effects in speech errors. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44(44), 3411–3417.
- Stern, M. C., & Shaw, J. A. (2023). Neural inhibition during speech planning contributes to contrastive hyperarticulation. *Journal of Memory and Language*, 132, 104443. <https://doi.org/10.1016/j.jml.2023.104443>
- Stevens, K. N., Liberman, A. M., Studdert-Kennedy, M., & Öhman, S. E. G. (1969). Crosslanguage study of vowel perception. *Language and Speech*, 12(1), 1–23. <https://doi.org/10.1177/002383096901200101>
- Thies, T., Mücke, D., Lowit, A., Kalbe, E., Steffen, J., & Barbe, M. T. (2020). Prominence marking in parkinsonian speech and its correlation with motor performance and cognitive abilities. *Neuropsychologia*, 137, 107306. <https://doi.org/10.1016/j.neuropsychologia.2019.107306>
- Tilsen, S. (2009). Subphonemic and cross-phonemic priming in vowel shadowing: Evidence for the involvement of exemplars in production. *Journal of Phonetics*, 37(3), 276–296. <https://doi.org/10.1016/j.wocn.2009.03.004>
- Van Ooijen, B. (1996). Vowel mutability and lexical selection in English: Evidence from a word reconstruction task. *Memory & Cognition*, 24(5), 573–583. <https://doi.org/10.3758/BF03201084>
- Wedel, A., Nelson, N., & Sharp, R. (2018). The phonetic specificity of contrastive hyperarticulation in natural speech. *Journal of Memory and Language*, 100, 61–88. <https://doi.org/10.1016/j.jml.2018.01.001>
- Wells, J. C. (1982). *Accents of english* (Vol. 1). Cambridge University Press.
- Whalen, D. H. (1990). Coarticulation is largely planned. *Journal of Phonetics*, 18(1), 3–35. [https://doi.org/10.1016/S0095-4470\(19\)30356-0](https://doi.org/10.1016/S0095-4470(19)30356-0)

Appendix

A.1. SVM accuracy results & sample plots

Below, we present the cross-validation scores from the SVM models used to classify participants' vowel tokens as errorful or non-errorful. Cross-validation is a measure of how well the model generalizes. K-fold cross-validation, a type of cross-validation technique, works by holding out different subsets of the training data, and computing the accuracy of the model created based on the remaining data. The number of subsets to be taken across the data is specified by the modeller, giving us the "k" in K-fold cross-validation; ours was 10, meaning we carried out a 10-fold cross validation. The mean of these scores across the different *folds* can tell us if the model generalizes well or if it depends too much on the training data. High scores indicate better performance. Based on Table A1, we can see that the models are performing well and we can be confident that the vowel classification will work as expected. The figures below the table represents some sample figures (Figures A1–A2) illustrating how our two-class SVM works. The line represents the decision boundary separating the two classes, while the individual points represent each token. All SVM plots can be found on the OSF: DOI 10.17605/OSF.IO/X8AKQ.

Table A1. SVM cross-validation scores.

Vowel pair	Mean score
/æ/ - /ɑ/	100%
/ɪ/ - /ɛ/	100%
/i/ - /u/	100%
/ʊ/ - /ʌ/	94.4%

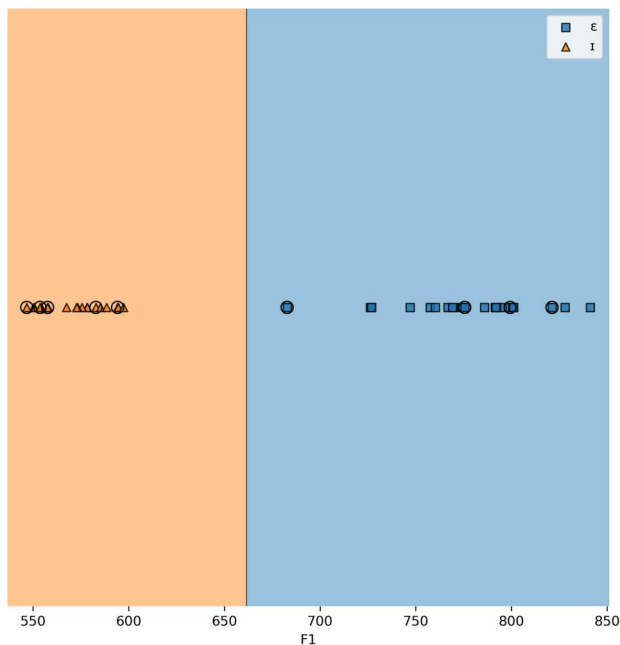


Figure A1. SVM sample plot for vowels /ɪ/ - /ɛ/.

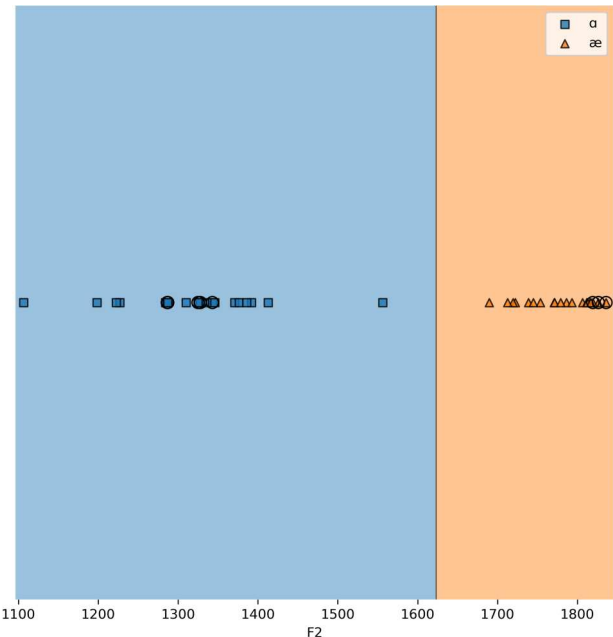


Figure A2. SVM sample plot for vowels /æ/ - /ɑ/.

A.2. Data-based plots

Below we present raw data plots for formants (Figure A3–A4) and duration (Figure A5–A6).

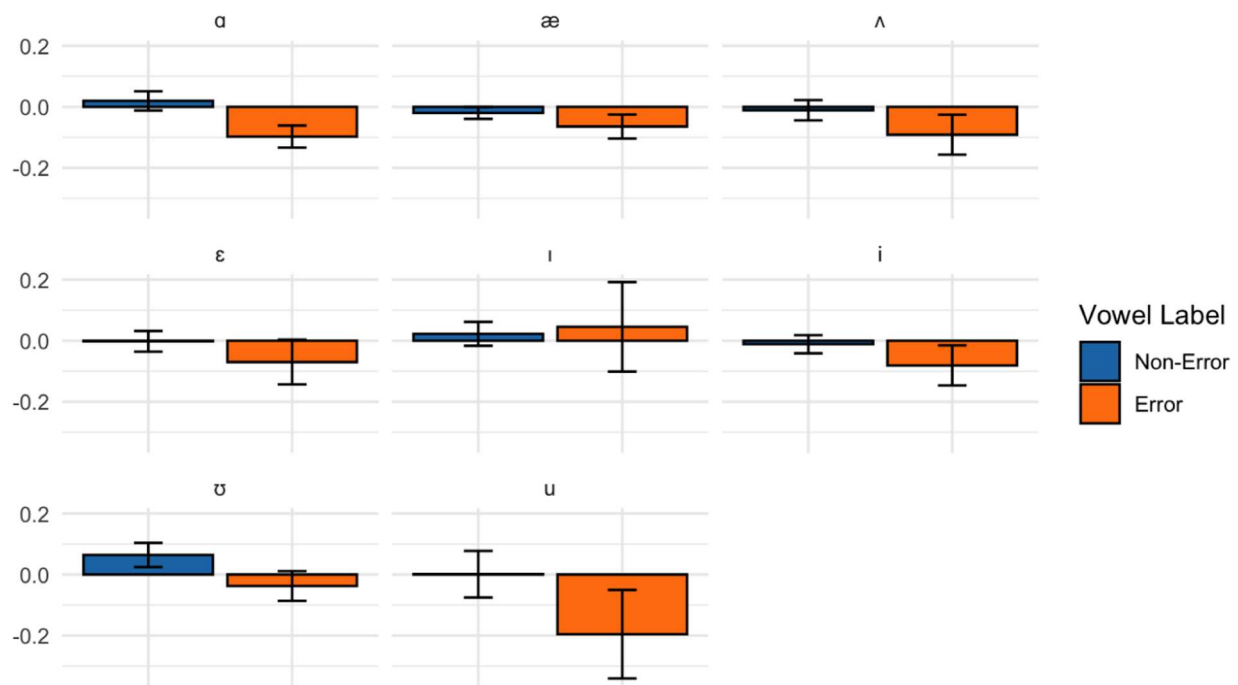


Figure A3. Formant log ratio means for errors and non-errors by vowel. Error bars represent 1 standard error.

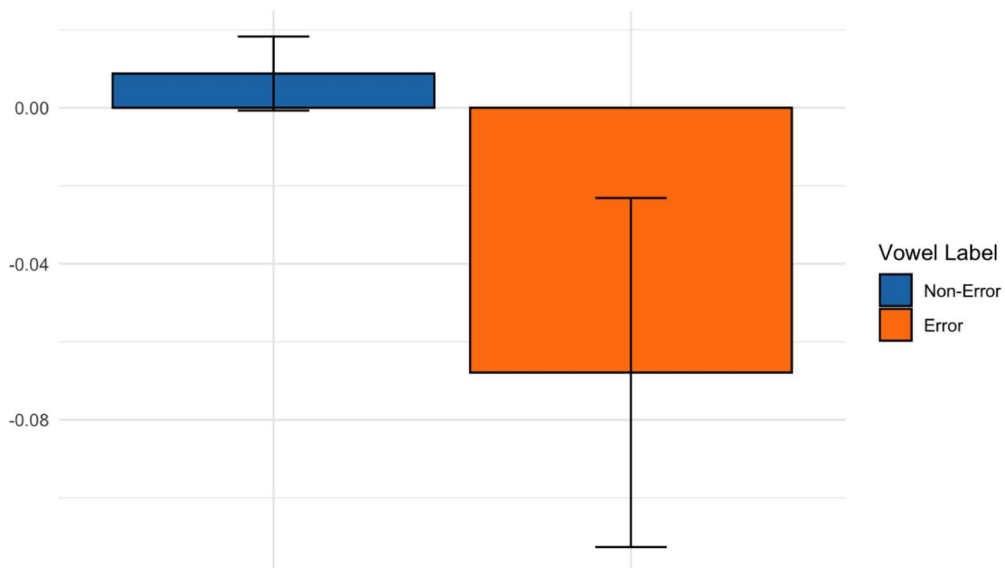


Figure A4. Formant log ratio means for errors and non-errors across vowels. Error bars represent 1 standard error.

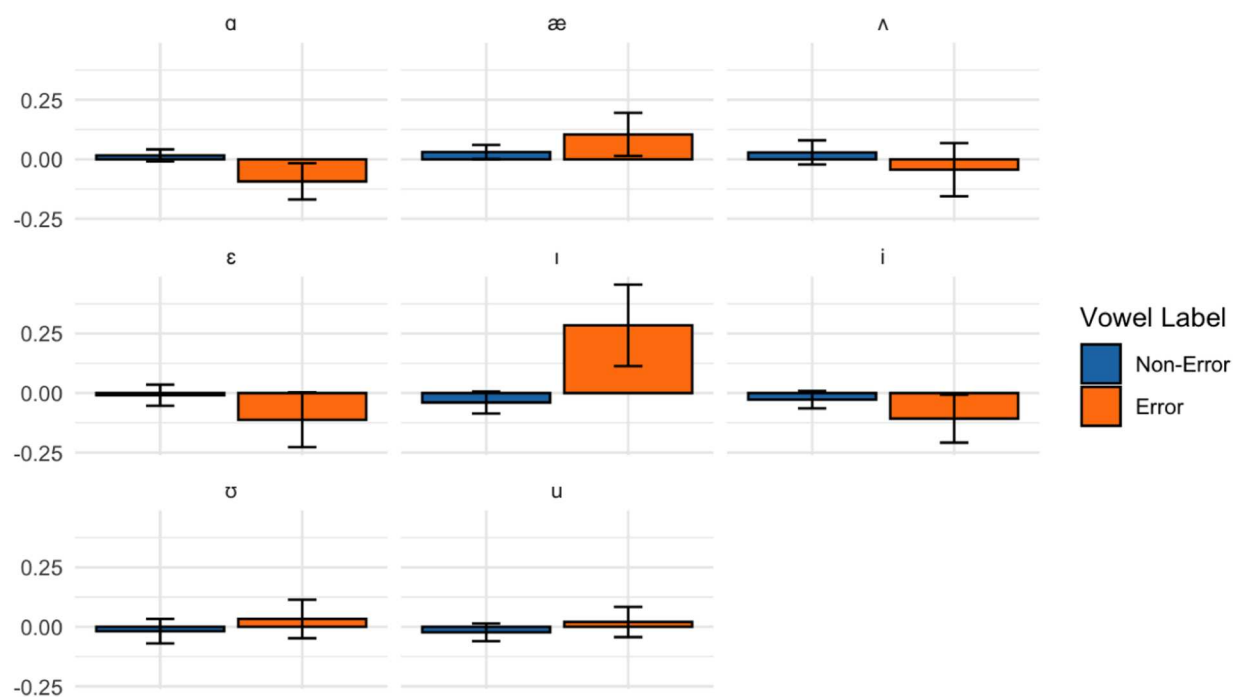


Figure A5. Duration log ratio means for error and non-error by vowel. Error bars represent 1 standard error.

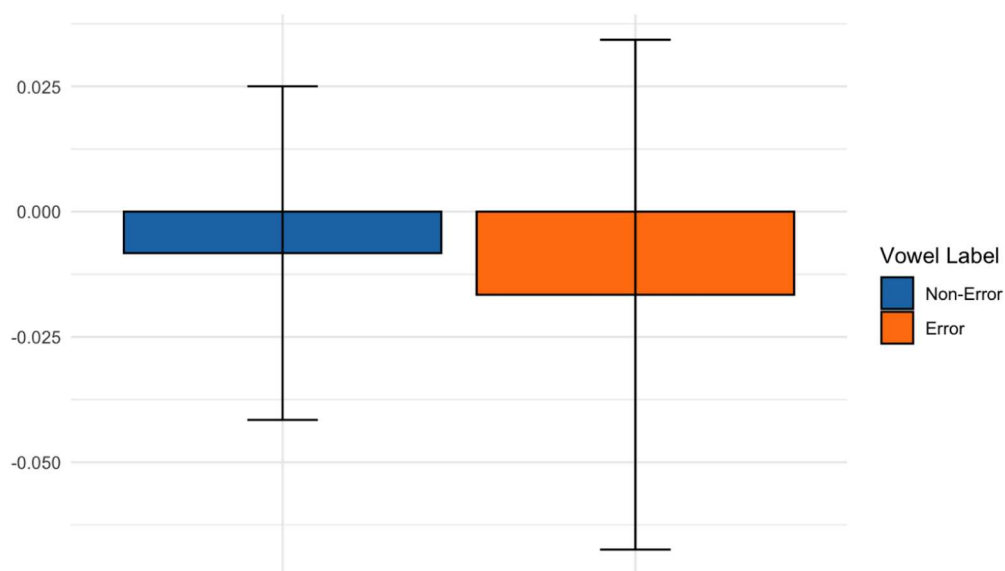


Figure A6. Duration log ratio means for error and non-error across vowels. Error bars represent 1 standard error.